

Ritesh Kumar

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Professional Summary

Software Engineer with experience in AI-driven systems and scalable backend development. Built deep learning models for industrial defect detection and developed full-stack applications with RESTful APIs and database integration. Focused on delivering efficient, real-world solutions through data-driven and system-oriented approaches.

Education

Arka Jain University, Jamshedpur

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.47

Aug. 2022 – May 2026

Experience

Machine Vision and Intelligence Lab, NIT Jamshedpur

Research Intern

May 2025 – Present

Jamshedpur (On-Site)

- **Steel Surface Defect Detection** — Developed a lightweight steel surface defect detection model using MobileNetV2, Feature Pyramid Network (FPN), and attention-based multi-scale feature fusion on the NEU-DET dataset.
- **Model Performance** — Achieved 98.33% classification accuracy and 0.85 mAP, optimizing the architecture for real-time edge deployment in industrial manufacturing environments.

Web Development Intern

TechMantra Global

May 2024 – July 2024

Noida (Remote)

- **Backend & API Development** — Developed a Food & Recipe API website using Node.js and MySQL, implementing 10+ RESTful APIs that reduced data retrieval latency by approximately 35% and optimized database query performance.
- **Tech Stack & Technologies** — Worked with HTML, CSS, Bootstrap, JavaScript, MySQL, and RESTful APIs for full-stack web application development and backend integration.

Projects

SoEIT Student Achievement Portal

Live | [GitHub Link](#)

- **Achievement Management System** — Built a full-stack academic platform using React.js and Node.js to track achievements for 1000+ students, integrating JWT-based role authentication to securely manage student and faculty dashboards.
- **Secure Authentication** — Implemented JWT-based authentication with role-based access control for students, faculty, and administrators to ensure secure and controlled system usage.
- **Tech Stack** — Node.js, Express.js, React, React Native, Turso (LibSQL), JWT, REST APIs.

AI-Powered Customer Complaint Management System

Live | [GitHub Link](#)

- **Complaint Management System** — Developed a secure web-based system using FastAPI and PostgreSQL with JWT authentication, enabling efficient complaint submission, tracking, and management through RESTful APIs.
- **Tech Stack** — FastAPI, PostgreSQL, JWT, REST APIs, Python, React, HTML, CSS, JavaScript.

Steel Surface Defect Detection

[GitHub Link](#)

- **Deep Learning Model Development** — Designed and trained an optimized CNN-based model to classify steel surface defects (crazing, inclusion, patches, pitted_surface, rolled-in_scale, scratches), leveraging multi-scale feature fusion and attention mechanisms to enhance detection accuracy in industrial inspection environments.
- **Keywords** — Deep Learning, MobileNetV2, Attention Mechanism

Publications

- **Cross-Enhanced and Attention-Modulated Multi-Scale Feature Fusion for Efficient Industrial Surface Defect Identification**

Ritesh Kumar, K. K. Singh — *Discover Artificial Intelligence*, Springer Nature (Under Review, 2026).

Technical Skills

Programming Languages: C/C++, Python, JavaScript, HTML

Frameworks & Libraries: React.js, Node.js, TensorFlow

Developer Tools: VS Code, Git, GitHub, Jupyter Notebook

Databases: MySQL, MongoDB

Core Technologies: Machine Learning, Deep Learning, REST APIs

Coursework: Data Structures and Algorithms, Object-Oriented Programming, DBMS

Achievements

- **AutonomousHacks '26** — Shortlisted for the offline finale among 2,000+ participants; selected for an Agentic AI hackathon organized by GDG Gandhinagar.
- **IBM Hackathon 2025** — National Finalist; qualified through IBM HackerRank challenge and competed in a 24-hour national hackathon.
- **CodeFest, Arka Jain University** — Secured **2nd** place in the Engineering Day coding competition.
- **NPTEL (Database Management Systems)** — Successfully completed certification.
- **Hack Horizon 2K25** — Organized and coordinated a university-level hackathon, promoting innovation and collaboration.

Leadership & Extracurricular Activities

- **Club President, Code & Compute Society (CCS), Arka Jain University** — Led operations, coordinated with faculty, and drove coding initiatives.
- **Community Lead, GDG on Campus AJU** — Organized workshops, events, and peer-learning sessions.
- **Class Representative (CR)** — Bridged communication between faculty and students.
- **Team Member, Gate Club, Arka Jain University** — Supported GATE preparation through workshops and discussions.